Code No:111AE **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY, HYDERABAD
B.Tech I Year Examinations, December – 2018
ENGINEERING CHEMISTRY
(Common to CE, EEE, ME, ECE, CSE, EIE, IT, MCT, ETM, MMT, AE, MIE, PTM, AGE)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART - A

(25 Marks)

1.a) What is molar conductivity ? Explain the temperature effect on molar conductivity.[2]
b) What are the cathode, anode and electrolyte in Ni-Cd cell. [3]
c) What are the conducting polymers and give two examples. [2]
d) Write any two applications of nanomaterials . [3]
e) Why is NH_3-NH_4Cl buffer solution added during determination of headness of water by EDTA method? [2]
f) What is reverse osmosis? Explain. [3]
g) Why should an ideal fuel have moderate ignition temperature? [2]
h) Why is net calorific value less than gross calorific value? Give reason. [3]
i) Define phase role. [2]
j) Explain classification of colloids. [3]

PART - B

(50 Marks)

2.a) Explain construction and working of Calomel electrode.
b) Explain the electrochemical theory of corrosion. [5+5]

OR

3.a) Explain construction working and applications of glass electrode.
b) What are fuel cells ? How methanol fuel cell works and writes its applications. [5+5]

4.a) Classify the following into plastics, elastomers and fibers and write their preparation with chemical equation.
i) PVC ii) Nylon iii) BUNa - S
b) What is vulcanization of rubber ? Write its uses. [5+5]

OR

5.a) Explain preparation, properties and applications of Bakelite.
b) Explain the concept of Biodegradable polymers ? Write preparation and applications of polylactic acid. [5+5]

6.a) What are the water quality parameters ? Write their significance.

b) What are Boiler troubles? Name them and explain the internal treatment methods of boiler feed water. [5+5]

OR

7.a) What is meant by sterilization of water ? Explain how sterilization of water is carried out by using chlorine and ozone.

b) How to estimate hardness of water by EDTA method. [5+5]

8.a) What is meant by knocking ? How is it related to chemical constitution ? describe the functions of TEL.

b) Differentiate proximate and ultimate analysis of coal. [5+5]

OR

9.a) What is synthetic petrol ? With neat diagram and chemical equations how synthetic petrol is prepared by Fischer-Tropsch's process.

b) Explain the composition, properties and applications of LPG and CNG. [5+5]

10.a) Explain phase diagram of head-silver system.

b) What are the applications of colloids in industry? [5+5]

OR

11.a) Explain phase diagram of water system.

b) What are the applications of adsorption? [5+5]

---ooOoo---